IoT-Based Home Automation System Using ESP32 and Blynk

1. Introduction

The Internet of Things (IoT) is one of the fastest-growing technologies in modern automation systems. IoT enables electronic devices to communicate over the internet, allowing remote monitoring and control of systems from anywhere in the world. One of the major applications of IoT is home automation, where electrical appliances such as lights, fans, and other devices can be controlled wirelessly using smartphones or cloud platforms.

This project presents the development of an IoT-based home automation system using an ESP32 microcontroller, relay modules, LEDs, and the Blynk platform. The purpose of the project was to simulate a smart home lighting system in which lights can be switched ON and OFF remotely through internet connectivity.

In this system, the ESP32 acts as the main controller because of its built-in Wi-Fi capability. Four relay modules were connected with the ESP32, and each relay controlled an LED representing a home light. The Blynk platform was used as the remote interface to send commands to the ESP32 through Wi-Fi.

The project demonstrates practical implementation of embedded systems, wireless communication, cloud connectivity, and automation techniques used in modern smart homes.

2. Objectives

The main objectives of this project were:

- To design a low-cost IoT-based home automation system.
- To interface relay modules with the ESP32 microcontroller.
- To remotely control LEDs using the Blynk IoT platform.
- To establish wireless communication using Wi-Fi technology.
- To understand the practical implementation of IoT in home automation systems.

3. Components Used

Component	Quantity	Description
ESP32 Development Board	1	Main microcontroller with Wi-Fi capability
Relay Modules	4	Used for switching loads
LEDs	4	Simulated home lights
Jumper Wires	Multiple	Electrical connections
Blynk IoT Platform	1	Remote control interface
Arduino IDE	1	Programming environment

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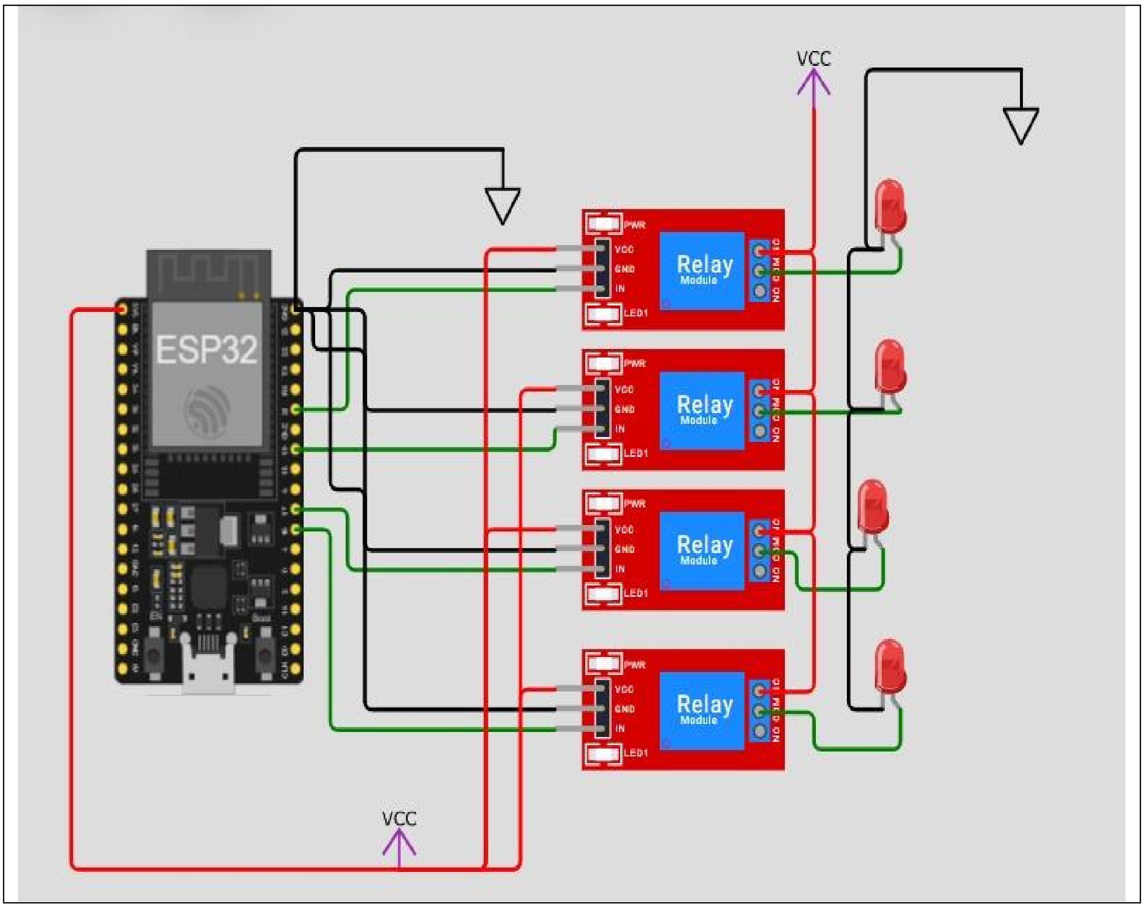
4. Circuit Diagram Description

The circuit consists of an ESP32 connected to four relay modules. The GPIO pins of the ESP32 are connected to the input pins of the relay modules. The relay module receives control signals from the ESP32 and switches the connected LED accordingly.



The following GPIO pins were used:

ESP32 Pin	Connected Device
GPIO 16	Relay 1
GPIO 17	Relay 2
GPIO 19	Relay 3
GPIO 21	Relay 4

Each relay module contains three important terminals:

- COM (Common)
- NC (Normally Closed)
- NO (Normally Open)

In this project, the Normally Closed (NC) terminal of each relay was connected to the positive terminal of an LED. The negative terminals of the LEDs were connected to ground.

The relay modules and ESP32 were powered through a common VCC and GND connection.

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5. Working Principle

The working of the system is based on wireless communication between the Blynk cloud server and the ESP32 microcontroller.

Initially, the ESP32 connects to a Wi-Fi network using the SSID and password programmed into the Arduino code. After establishing internet connectivity, the ESP32 connects to the Blynk cloud server using the authentication token provided by the Blynk platform.

The Blynk mobile application contains virtual buttons assigned to virtual pins V0, V1, V2, and V3. When a user presses a button in the Blynk interface, a signal is transmitted over the internet to the ESP32.

The ESP32 receives the command and executes the corresponding `BLYNK_WRITE()` function. The microcontroller then changes the state of the associated GPIO pin, which activates or deactivates the relay module. As a result, the connected LED turns ON or OFF.

This process demonstrates how household lighting systems can be remotely controlled using IoT technology.

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6. Software Implementation

The software was developed using the Arduino IDE with ESP32 board support and Blynk libraries installed.

The following libraries were included in the code:

```
#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
```

These libraries enabled Wi-Fi communication and interaction with the Blynk platform.

The Blynk authentication token, template ID, Wi-Fi SSID, and password were defined in the code to establish communication between the ESP32 and the Blynk cloud server.

The GPIO pins connected to the relays were defined as output pins:

```
#define LED1 16
#define LED2 17
#define LED3 19
#define LED4 21
```

Inside the `setup()` function, all relay pins were configured as outputs and the Blynk connection was initialized.

The `loop()` function continuously executed:

```
Blynk.run();
```

This maintained communication between the ESP32 and the Blynk server.

The following functions controlled the relay states:

```
BLYNK_WRITE(V0)
BLYNK_WRITE(V1)
BLYNK_WRITE(V2)
BLYNK_WRITE(V3)
```

Each function corresponded to a virtual button in the Blynk application and controlled a specific relay module.

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7. Advantages of the System

The developed system offers several advantages:

- Remote control of devices through internet connectivity.
- Low-cost implementation using ESP32.
- Easy user interface using Blynk.
- Expandable design for future automation projects.
- Reduced manual operation of electrical appliances.
- Real-time wireless control capability.

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8. Conclusion

The IoT-based home automation project successfully demonstrated wireless control of lighting devices using the ESP32 microcontroller and Blynk platform. The system allowed LEDs, representing household lights, to be controlled remotely through internet communication.

The ESP32 provided efficient Wi-Fi connectivity while the relay modules enabled switching of external loads. The Blynk platform offered a simple and user-friendly interface for remote operation.

This project provided practical experience in IoT systems, embedded programming, relay interfacing, and cloud-based communication. It also demonstrated the importance of smart automation technologies in modern homes and industries.

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9. References

- ESP32 Official Documentation
- Arduino IDE Official Website
- Blynk IoT Platform